Midterm Sample 1

ECON 441: Introduction to Mathematical Economics Instructor: Div Bhagia

Print Name: ______________________________

This is a closed-book test. You may not use a phone or a computer.

Time allotted: 70 minutes
Total points: 20

Please show sufficient work so that the instructor can follow your work.

I understand and will uphold the ideals of academic honesty as stated in the honor code.

Signature: ____________________

1. (4.5 pts) Answer the following questions

   (a) (1 pt) Consider two sets A and B, where A is the set of all odd real numbers and B is the set of all even real numbers. What is the union of A and B?

   (b) (1 pt) Expand the following summation expression:

   $$\sum_{j=1}^{2} \sum_{i=1}^{2} X_i Y_j =$$

   (c) (1pt) Does the inverse of the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$ exist? If yes, find the inverse.

(d) (1.5 pts) Find $A^T A$ where

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Remember: A^T is the transpose of A.

2. (8 pts) Two types of cars, gasoline (g) and electric (e), are available in the market. Denote the price of gasoline cars by p_g and the price of electric cars by p_e. Similarly, denote the quantity of gasoline cars as q_g and electric cars as q_e. The supply and demand equations for these cars are as follows:

   Supply Equations:

$$q_e^s = 25 + 0.3p_e$$
$$q_g^s = 50 + 0.3p_g$$

   Demand Equations:

$$q_e^d = 100 - 0.5p_e + 0.2p_g$$
$$q_g^d = 150 + 0.2p_e - 0.5p_g$$

   Note: At equilibrium, supply is equal to demand: $q_e^s = q_e^d = q_e$ and $q_g^s = q_g^d = q_g$.

   (a) (2 pts) Write down the four equations that must hold in equilibrium. Rearrange each equation so that all terms with variables are on one side and all constants are on the other. What are the four unknown variables?

(b) (2 pts) Express this system of equations in matrix format as $Ax = b$. Clearly specify what A, x, and b are.

(c) (2 pts) What is the necessary and sufficient condition for a unique solution for this system of equations to exist?

(d) (2 pts) How would you solve this system of equations using tools from matrix algebra? There's no need to actually solve it; just explain the steps involved.